

IDMERS:

Integrated Flood and Landslide Risk Assessment and Response System for Sri Lanka

Predict. Warn. Respond.

Integrated Flood and Landslide
Risk Assessment and Response System for Sri Lanka

PROJECT ID: J26-IT-334



Recurring Risks. Real Impact.

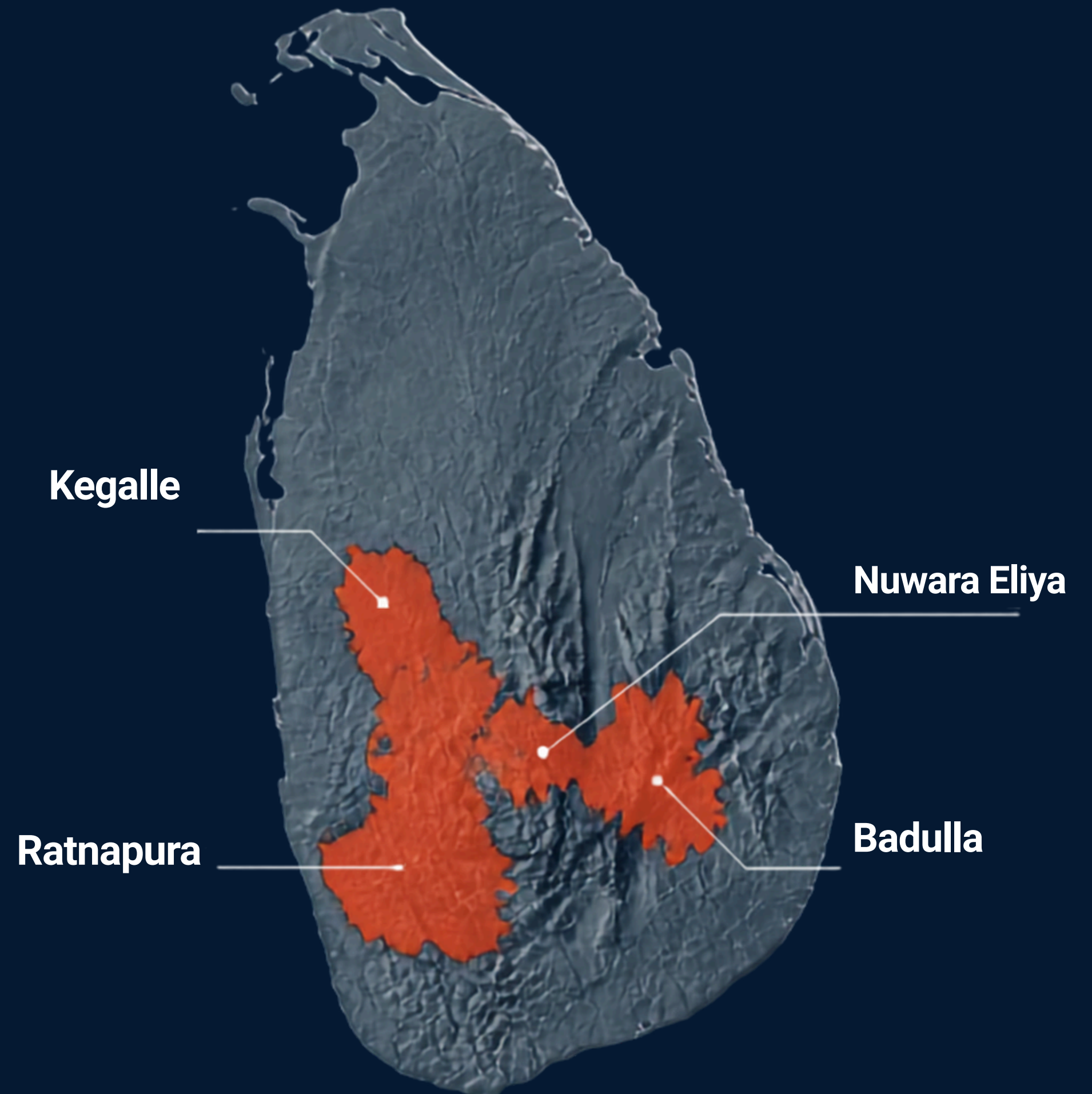
Floods and landslides continue to devastate communities across Sri Lanka.

Loss of life, **infrastructure damage**, **transport disruption**, and significant **economic losses**

High-risk areas: **Ratnapura, Kegalle, Badulla, Nuwara Eliya**

Driven by monsoon rainfall, urbanization, deforestation, land use changes, and climate variability

Current warning systems are separate, hazard-specific, and reactive

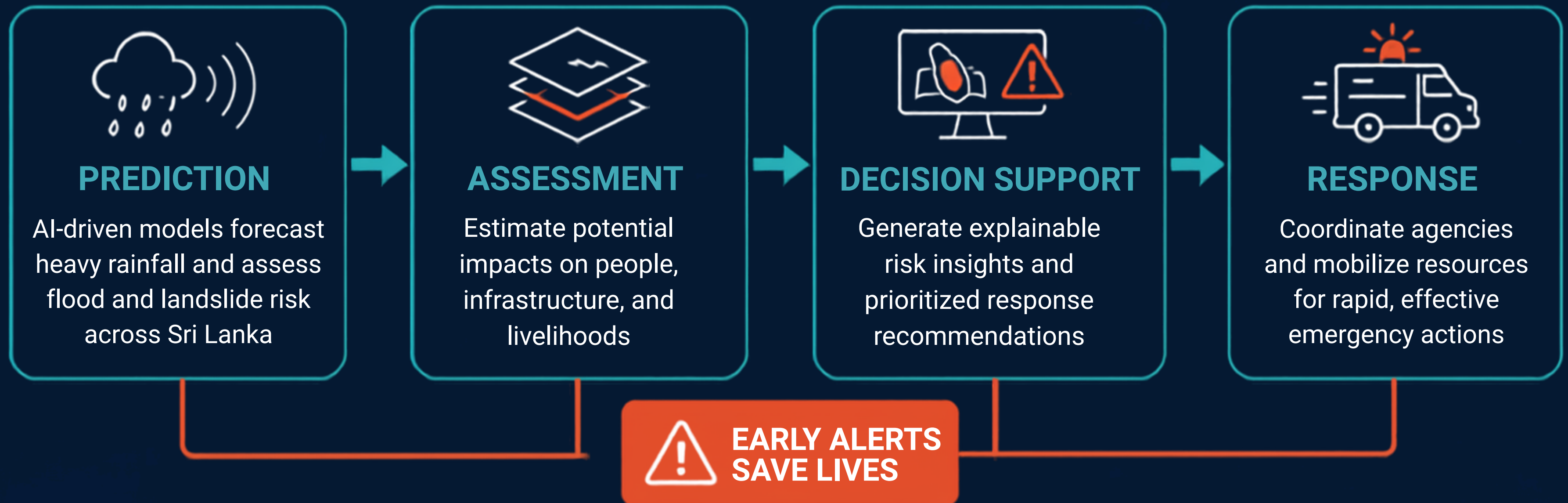


One System. Smarter Response.

An integrated, AI-driven system for flood and landslide risk

Four components working together from prediction to response

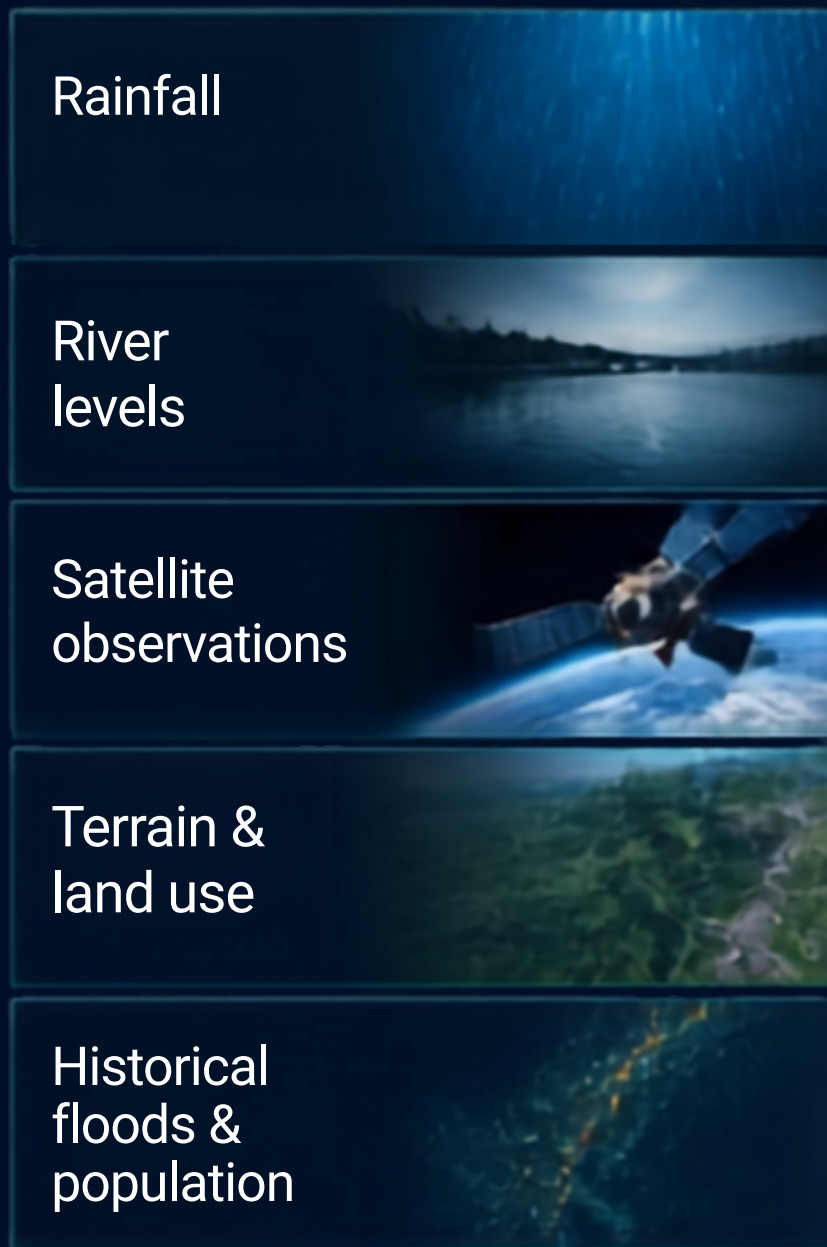
Timely, explainable, inclusive alerts and prioritized actions for Sri Lanka



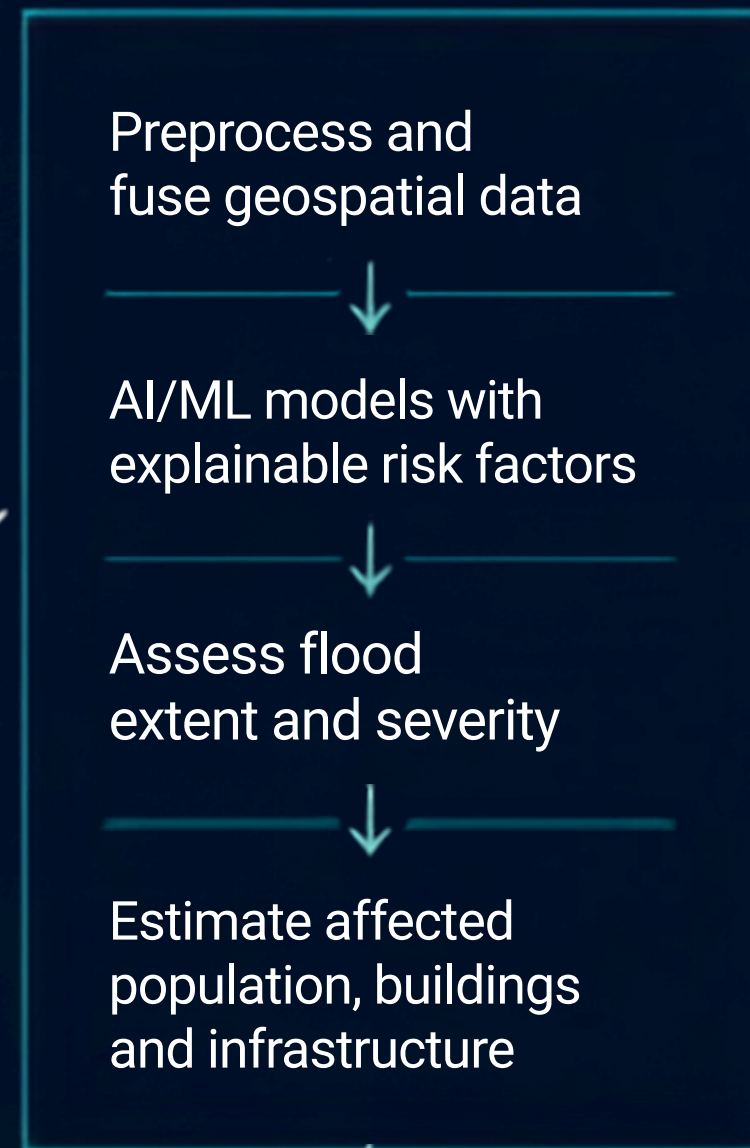
Component 1

Flood Severity Assessment & Mapping

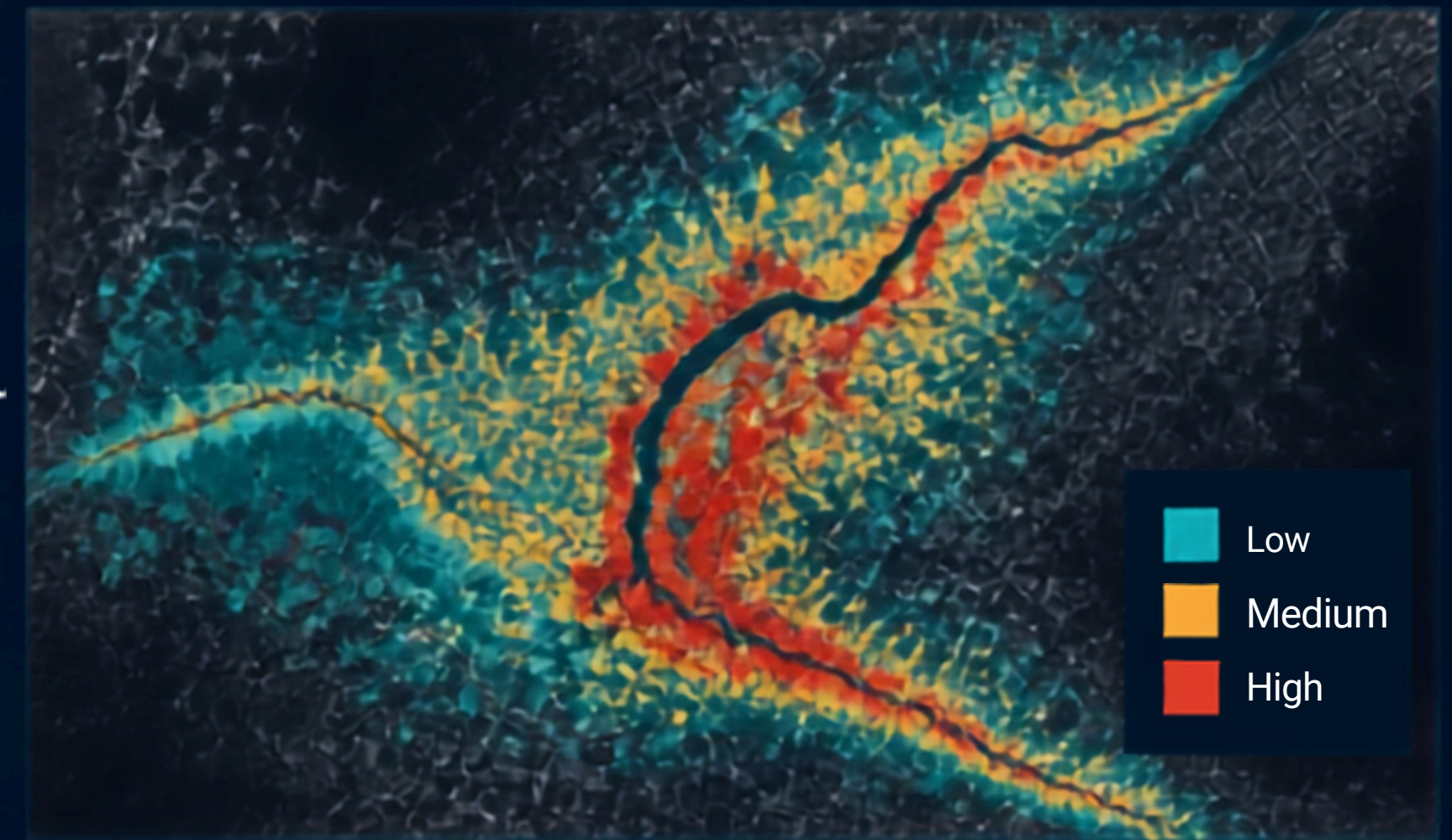
MULTI-SOURCE DATA INPUTS



PROCESSING & ANALYTICS



DYNAMIC FLOOD SEVERITY MAP



**Actionable insight
for response**

Supports evacuation planning
Enables resource prioritization
Guides decision-making in real time

Probabilistic Landslide Assessment & Forecasting

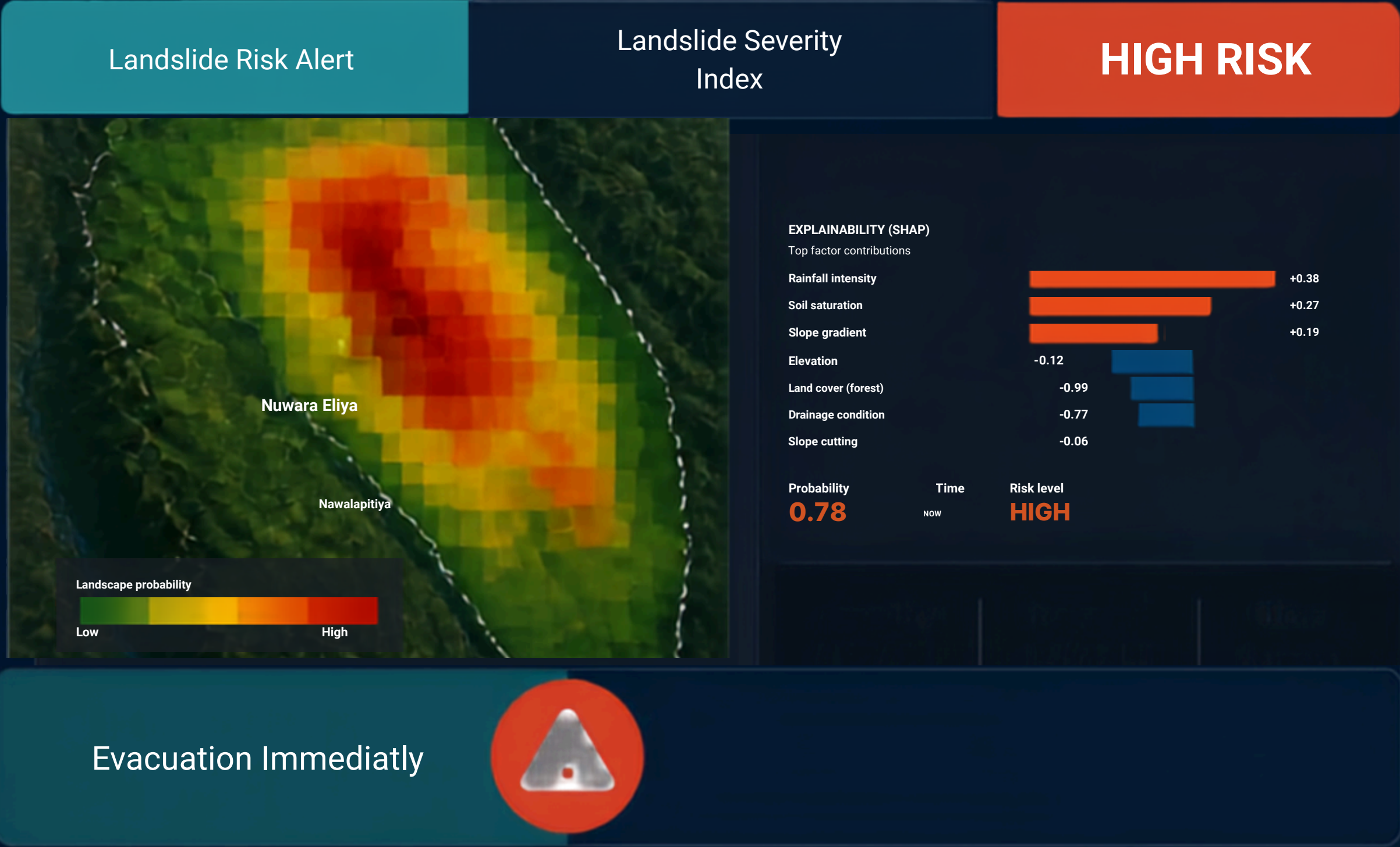
Machine learning and explainability with accessible alerts

Real-time landslide probability
from provincial to village scale

Explainable AI reveals the key
environmental and human drivers

Accessible Web-GIS and 3D maps with
Sinhala/Tamil dual-language alerts

Powered by Random Forest/XGBoost,
PostGIS, Leaflet/MapLibre, WebGL



Component 3

Human Impact & Decision Support.

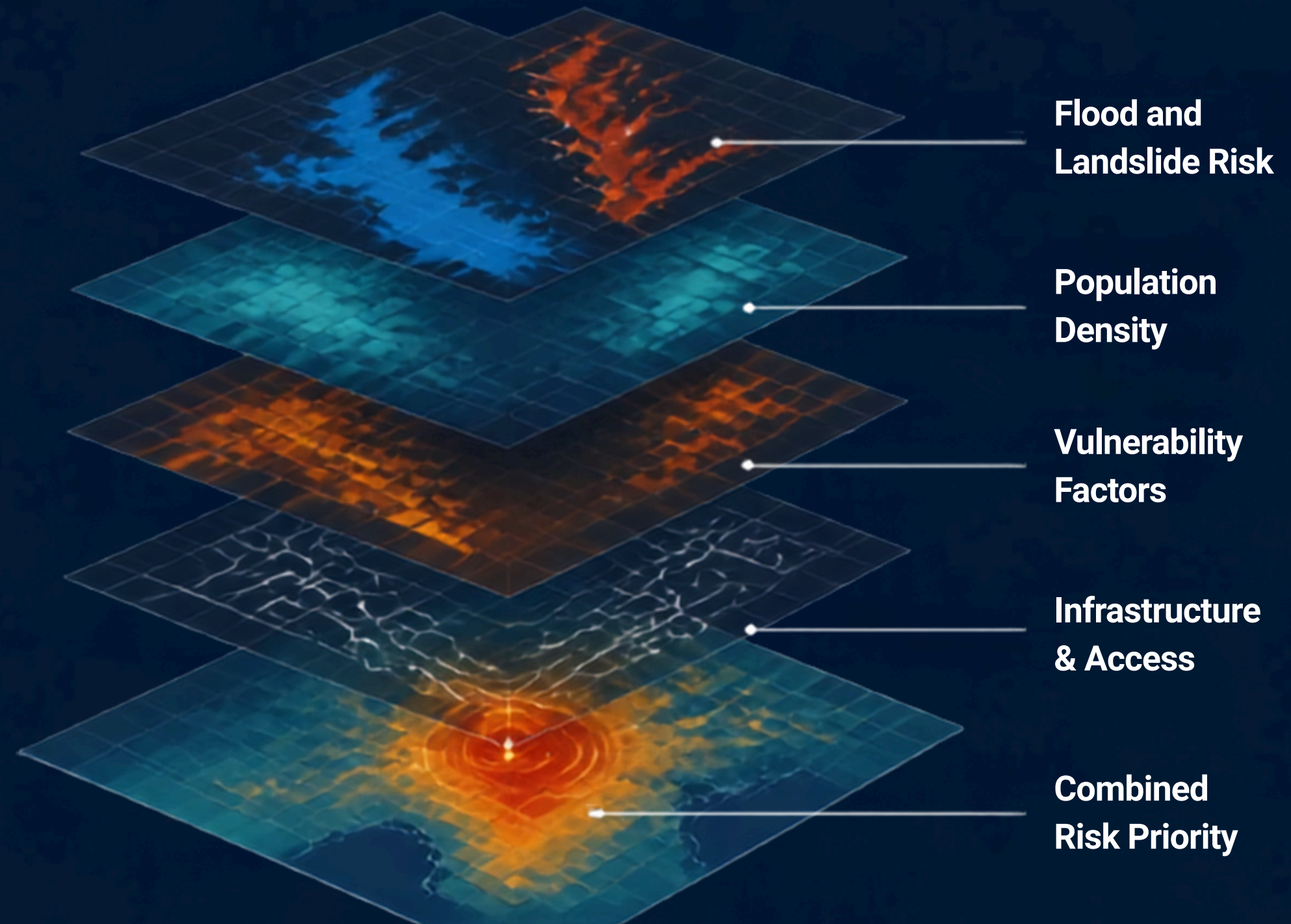
Brings hazard data together with population and social vulnerability

Considers critical infrastructure, access, history, uncertainty, and temporal overlap

Models joint and cascade risks for deeper insight

Produces clear priority scores for fair, explainable decisions

Delivers graded alerts, actions, and prioritized response zones



HIGH PRIORITY

ALERT

Immediate Action
Recommended

MEDIUM PRIORITY

ALERT

Prepare and Monitor

LOW PRIORITY

ADVISORY

Stay Informed

Component 4

Emergency Response & Resource Coordination

Real-time resource-aware dispatch and routing innovations



INNOVATION

Real-time
resource-aware dispatch

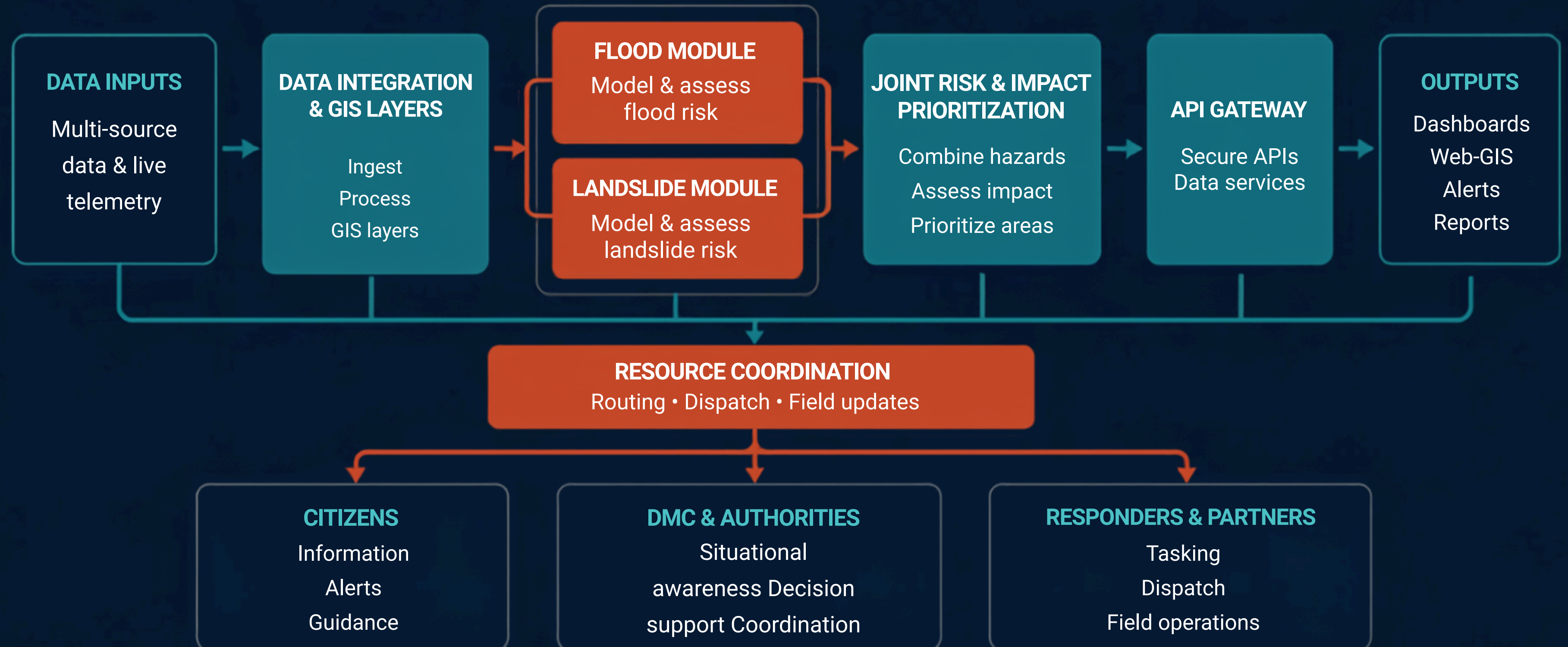
Dual-route
optimization

Scarcity-based
queuing

Multi-channel
localized dispatch

System Architecture.

One Integrated Workflow. End-to-End Impact.



Team and Supervisors.

OUR PROJECT TEAM

Sinhalagoda S.A.U.R.A.V (IT23280892)

Flood risk prediction & mapping

Bagya R M S (IT23394124)

ML landslide forecasting

Prathibhani Y.L.O.L (IT23317512)

Human impact & decision support

Palihena M.B (IT23411562)

Emergency response & resource coordination

ACADEMIC GUIDANCE

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Co-Supervisor

PROJECT ID

J26-IT-334

PROJECT TITLE

**IDMERS: Integrated Disaster
Management and Early Warning System**



Thank You

Predict. Warn. Respond.

Strengthening communities
with trusted risk intelligence and
coordinated action.

Let's work together.

Connect with us to explore partnerships,
share data, and build a safer Sri Lanka.

NBRO LIRIS | ReliefWeb/OCHA Sri Lanka floods & landslides | Dept. Meteorology Sri Lanka | DMC flood warning
UNDRR Sendai Framework | Palliyaguru et al. (ICTer 2020) | Atthanayake et al. (SLAAI-ICAI 2025)